

10. LIST OF APPENDICES

1. **Appendix A** – Execution steps

1. **Hardware Setup**

- Connect ESP32 to:
 - GPIO13, GPIO4, GPIO5, GPIO12 → Relay/Appliance Output
 - GPIO14, GPIO27, GPIO26, GPIO25 → Manual Push Buttons
- Interface LCD (I2C Address: 0x27, 16x2) with SDA and SCL pins.
- Connect Voltage & Current sensors to analog pins (GPIO35 for voltage, GPIO34 for current).
- Ensure a stable 5V power supply to ESP32 and peripherals.

2. **Library Installation**

Install the following libraries via Arduino IDE Library Manager:

- EmonLib – for energy monitoring
- LiquidCrystal_I2C – for LCD display
- WiFiManager – for dynamic WiFi setup
- PubSubClient – for MQTT communication

3. **Arduino IDE Configuration**

- Board: Select **ESP32 Dev Module**
- COM Port: Select appropriate port
- Upload Speed: Default or 115200 baud rate
- Compile and upload the code to the ESP32 using USB cable

4. **Initial WiFi Configuration**

- On first boot, ESP32 launches a WiFi hotspot named **“ESP32_AP”**
- Connect your phone/laptop to this hotspot
- A portal opens automatically → Enter your local WiFi credentials
- ESP32 reboots and connects to your WiFi automatically

5. MQTT Integration

- HiveMQ Broker (broker.hivemq.com) used for switching devices remotely
 - Topics: switch11, switch12, switch13, switch14
- ESP32 subscribes to these topics and controls GPIOs accordingly

6. Button and Manual Override Control

- Each button toggles the corresponding GPIO relay
- Manual control is given priority to avoid MQTT override during physical interaction
- Button presses are debounced for stable switching

7. Energy Monitoring

- Voltage and current are calculated using EmonLib
- $\text{Power} = V_{\text{rms}} \times I_{\text{rms}}$

8. ThingSpeak Dashboard Integration

- MQTT credentials used to publish data to ThingSpeak:
 - Power consumption
 - GPIO states (0 or 1)
- Updates sent every 20 seconds to the configured ThingSpeak channel
- Enables remote monitoring and visualization of appliance states and energy usage

9. Multitasking Using FreeRTOS

- gpioControlTask() → Manages button inputs
- mqttTask() → Handles HiveMQ client loop
- energyMonitoringTask() → Calculates and displays power & energy
- thingSpeakTask() → Publishes data to ThingSpeak

10. Final Test

- Verify WiFi connection and MQTT subscription
- Test manual and remote switching
- Monitor values on LCD and ThingSpeak dashboard